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# Machine Learning Applied to Determine the Molecular Descriptors Responsible for the Viscosity Behavior of Concentrated Therapeutic Antibodies

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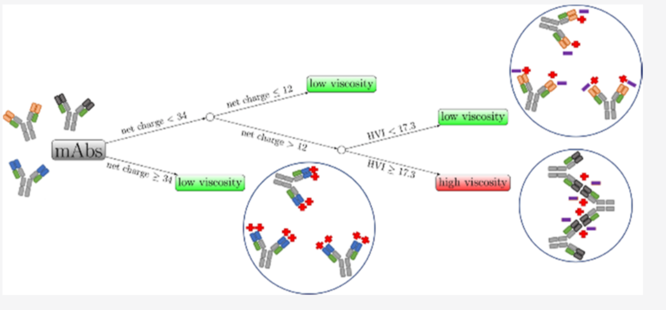
**ABSTRACT:** Predicting the solution viscosity of monoclonal antibody (mAb) drug products remains as one of the main challenges in antibody drug design, manufacturing, and delivery. In this work, the concentration-dependent solution viscosity of 27 FDA-approved mAbs was measured at pH 6.0 in 10 mM histidine-HCl. Six mAbs exhibited high viscosity (>30 cP) in solutions at 150 mg/mL mAb concentration. Combining molecular modeling and machine learning feature selection, we found that the net charge in the mAbs and the amino acid composition in the Fv region are key features which govern the viscosity behavior. For mAbs whose behavior was not dominated by charge effects, we observed that high viscosity is correlated with more hydrophilic and fewer hydrophobic residues in the Fv region. A predictive model based on the net charges of mAbs and a high viscosity index is presented as a fast screening tool for classifying low- and high-viscosity mAbs.




**KEYWORDS:** *therapeutic antibodies, viscosity, machine learning, molecular modeling, intermolecular interactions*



## 1. INTRODUCTION

Monoclonal antibodies (mAbs) have become a dominant class of therapeutic[1] and have fundamentally changed how drugs are administered to patients. While small-molecule drugs are predominantly delivered orally, mAbs must be administered parenterally in order to overcome their poor oral bioavailability. Among the parenteral routes, subcutaneous injection of mAb drug products provides a less-invasive and more-convenient option than intravenous infusion.[2] This is of higher importance for chronic indications where a product that can be self-administered in a home setting enables better patient adherence and compliance. Such patient-centric dosage forms are needed to realize the full therapeutic potential of mAbs. However, administration by this route aims toward lower volumes, which necessitates the development of highly concentrated mAb products.

Despite the need for patient-centric dosage forms, there are significant challenges in the manufacturing and formulation of high-concentration mAb products.[3] In particular, mAb solutions can exhibit a significant increase in solution viscosity with increasing concentration,[4,5] complicating manufacturing processes and limiting options for device-mediated delivery. Unfortunately, the underlying molecular contributors to the rheological behavior of mAbs are still poorly understood, making early identification and characterization of well-behaved mAbs challenging.

The electroviscous effect that describes the viscosity behavior by the net charge and charge distribution on the surface has been proposed to model mAb concentration−viscosity behavior.[6] Under typical formulation conditions (pH = 5.2 to 6.3), the constant region of IgG1 mAbs is predicted based on p$K_a$ of individual amino acids to carry a net positive charge, leading to strong repulsion with the constant region of other mAbs.[7] The mAb variable domain (Fv) also contributes to the net charge in a manner dependent on its composition. Based on the electroviscous effect, a more acidic Fv is predicted to predispose increased viscosity via reduced electrostatic repulsion. Indeed, repulsive interactions measured in a dilute solution are a hallmark of well-behaved mAbs.[8−14] In accordance with the electroviscous effect, the solution charge has been shown to contribute significantly to these interactions, but measurement of the interactions themselves

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provides a more complete linkage to viscosity than does the solution charge itself.[8]

A recent analysis of 59 mAbs demonstrated that two opposing molecular mechanisms could be triggered by reversible self-association (also called weak protein−protein attractions, attractive PPI, or reversible cluster formation) at high mAb concentrations. Among a large dataset of measured and computed biophysical properties, self-interactions showed a singularly effective ability to differentiate well-behaved mAbs from those that become highly opalescent or viscous at high concentration.[14] While the electrostatic repulsion implicit in the electroviscous effect is of importance, dilute solution interactions reflect additional intermolecular attraction arising from anisotropic charge distribution and the hydrophobic and the hydrophilic effects.[13,15] These lead to increased attractive PPI, and thus, high solution viscosity, at high mAb concentrations. These discussions highlight the complex nature of high solution viscosity and the multifactorial nature of the underlying molecular forces that govern this phenomenon.

Critical to the development of patient-centric dosage forms is the identification and selection of well-behaved mAbs from the varied pool of potential candidates produced during the discovery campaign. Since experimental measurements of viscosity at high concentrations are not feasible in early stages due to insufficient material availability, designing a fast screening tool for detecting potential high-viscosity mAbs is of paramount importance. Due to their advantages in throughput and sample consumption, computational methods have been of considerable interest. For instance, the spatial charge map (SCM) score, which calculates a score based on the negative charge distribution of exposed residues in the Fv region of mAbs, has successfully classified the high- and low-viscosity behaviors of 19 IgG1 mAbs from three different sources.[16] Additionally, an Fv region-based qualitative screening profile was developed to flag mAb candidates based on net charge, zeta potential, and pI of the Fv region.[17] Based on the results, a regression model was proposed to quantitatively predict mAb viscosity based on the pI of the Fv region.[10] Sharma et al. also proposed a predictive model that only depends on the sequence and empirical parameters of the Fv region. Additionally, Tomar et al. proposed an equation that uses the charges on the heavy- and light-chain variable regions and the charges on the hinge region and the hydrophobic surface area of the full-length antibody to predict the concentration-dependent viscosity curves.[18] In addition, some coarse-grained simulations have also been carried out to study the self-association of mAbs and the relationship of PPI with viscosity.[19−21]

In this work, we studied 27 mAb drugs approved by the Food and Drug Administration (FDA). The concentration dependence of viscosity from 50 to 200 mg/mL was measured at pH 6.0 in 10 mM histidine-HCl buffer. First, we separately attempted to use the predicted net charge and the SCM score to classify the experimental viscosity data. There were a few misclassifications from both approaches. Given that machine learning presents a powerful way to classify data, we applied it to search for molecular descriptors for those that accurately separated high and low viscosity. In addition, we also proposed a high viscosity index (HVI) as a metric for rapid screening of low- and high-viscosity mAbs. A decision tree model based on the mAb net charge and HVI was proposed. We evaluated the predictivity of this model using 38 antibody mutants reported by Apgar et al.[22] recently. In the end, the role of hydrophobic and hydrophilic effects on antibody viscosity is discussed to provide more insights into the mechanism.

## 2. MATERIALS AND METHODS

**2.1. Protein Preparation.** Twenty-seven mAbs were purchased from pharmacy vendors, consisting of 21 IgG1, 4 IgG2, and 2 IgG4 isotypes (Table S1). The sequences are available in the Supporting Information. The surfactant was removed from the drug product using DetergentOUT Tween spin columns (G-Biosciences), and the samples were exchanged into a buffer of 10 mM histidine hydrochloride, pH 6.0, for high concentration testing. Exchange was conducted by exhaustive dialysis into the surfactant-free formulation buffer by four successive dialysis cycles wherein the protein was dialyzed against a total of 8 L of buffer using 20 kDa MWCO Slide-A-Lyzer cassettes (Thermo Scientific). Following exchange, the samples were concentrated to a target of 220 mg/mL and diluted with surfactant-free formulation buffer to 200, 175, 150, 120, 100, and 50 mg/mL concentrations. Prior to measuring viscosity, samples were passed through 5 $\mu$m microcentrifuge filters (Merck Millipore), and concentrations were measured using a SoloVPE attachment (C Technologies) on a Cary 50 spectrophotometer.

**2.2. Viscosity Measurements.** The viscosity measurements used the methods as previously published.[14] Each measurement required 50 $\mu$L of the sample to measure the viscosity at 20 °C. No sample showed shear thinning behavior in the initial measurement at 10 °C, so we consider non-Newtonian effects to be negligible in these measurements. The experimental uncertainties in these experiments are ±0.1 °C on temperature and ±5% on viscosity. The concentration−viscosity data were reduced by fitting to a sample exponential function as applied elsewhere[14]

$$\eta = A \times \exp(kC) \tag{1}$$

here $C$ is the mAb concentration and $A$ and $k$ are the fitting parameters. Because there are some concentration differences in the sample preparation (Table S2), using the best fit parameters in eq 1 allowed the viscosity to be interpolated at 150 mg/mL.

**2.3. Computational Modeling of mAbs.** The mAb molecules were constructed by following the protocol proposed by Brandt et al.[23] Briefly, the structure of the Fab region was superimposed on a template structure of full IgG1, IgG2, and IgG4 models. The IgG1 template was obtained from the KOL/Padlan structure.[24,25] For the IgG2 and IgG4 models, the Fc regions (PDB: 4HAF and 4C54, respectively) were superimposed on the KOL/Padlan IgG1 structure.[26,27] The Fab structures were retrieved from either available crystal structures or homology models built from RosettaAntibody.[28−30] The glycan structure was G0F.

**2.4. Molecular Dynamics Simulations.** Molecular dynamics simulations were performed using all-atom structures with an explicit solvent using the TIP3P water model.[31] Simulation boxes were set up using VMD to place a single antibody in a water box extending 12 Å beyond the protein surface.[32] Simulations were performed at 300 K and 1 atm in the *NPT* ensemble, using the NAMD software package and the CHARMM36m force field.[33−35] The system pH was set to 6.0 to match the experimental pH by adjusting the protonation states of histidine residues using the PROPKA3 protocol.[36] Electrostatic interactions were treated with the particle mesh Ewald method, and van der Waals interactions were calculated


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using a switching distance of 10 Å and a cutoff of 12 Å.[37] The integration time step was set to 2 fs. Each mAb system was pre-equilibrated for 10 ns, followed by 50 ns production runs.

**2.5. Predictive Models of Viscosity Behavior.** Four models were applied to predict the viscosity behavior of the 27 mAbs. The SCM calculates a score based on the negative charges of the exposed residues in the Fv region.[16] High SCM scores, above some threshold value, indicate high-viscosity mAbs. The other two models developed by Li et al.[17] and Sharma et al.[10] quantitatively predict mAb viscosities. In addition, the TAP: Therapeutic Antibody Profiler web server was utilized to predict the antibody viscosity.[38]

**2.6. Feature Selection for Machine Learning.** Features were selected based on charge, hydrophobicity, and hydrophilicity properties shown in Table 1. Most of these properties

**Table 1. List of Physical Properties That Are Related to Charge, Hydrophobicity, and Hydrophilicity[a]**

| properties | description |
|---|---|
| number of hydrophobic residues (*N*_phobic) | A, F, I, L, M, P, V, W |
| number of hydrophilic residues (*N*_philic) | S, T, N, Q, Y, K, R, H, D, E |
| net charge | calculated by PROPKA3 |
| isoelectric point (pI) | calculated by PROPKA3 |
| charge symmetric parameter (CSP) | product of heavy and light chain charge |
| hydrophobicity index (HI) | sequence-based hydrophobicity scale |
| solvent-accessible surface area of hydrophobic residues (SASA_phobic) | calculated by VMD |
| solvent-accessible surface area of hydrophilic residues (SASA_philic) | calculated by VMD |
| spatial charge map (SCM) | structure-based negative charge scale |
| spatial aggregation propensity (SAP) | structure-based hydrophobicity scale |

[a]These properties are calculated for the variable region and the full-length antibody. In total, there are 20 descriptors/parameters for feature selection.

were proposed to correlate with viscosity behavior.[10,16−18] We calculated these properties on both variable regions and full-length antibodies. In total, there are 20 features for selection (Table 1). The number of features (20) was specifically chosen to be less than the number of data points (27) to reduce the potential for overfitting.[39]

For the net charge calculation, it was assumed that arginine (+1), lysine (+1), aspartic acid (−1), and glutamic acid (−1) are fully charged at pH 6. The titration state of histidine was determined using PROPKA3.[12] The sequence-based net charge does not include the effect of solution ions.[40] The isoelectric point (pI) was also calculated from PROPKA3 using either the energy-minimized Fv or the full-length structures. The charge symmetric parameter (CSP) is a metric for charge asymmetry. This is the product of the total charge on the heavy chain and the total charge on the light chain. A negative CSP indicates high charge asymmetry. The hydrophobicity index (HI) follows the definition of Sharma et al.,[13] which is based on the Eisenberg hydrophobicity scales.[41] The spatial aggregation propensity is a measure of the surface-exposed hydrophobicity.[42] The complete list of feature values is available in the Supporting Information.

**2.7. Machine Learning Methods.** Logistic regression was used to train the machine learning models for feature selection and viscosity classification. Because of the limited amount of mAb viscosity data, we aimed to search for the smallest set of features that provided high accuracy. The exhaustive feature selector tool from mlxtend library was implemented to include all possible feature combinations.[43] The average area under the precision-recall curve (AUPRC) and average accuracy were evaluated with a threefold cross-validation approach. In addition, a decision tree model was also used for classifying low and high viscosities. A decision tree has an advantage for training nonlinear models and works well when features are on completely different scales.[44] Scikit-learn was used for model training and analysis.[39]

## 3. RESULTS AND DISCUSSION

**3.1. Concentration-Dependent Viscosity Behavior.** The concentration-dependent viscosity curves of all 27 mAbs and the results were used to interpolate the viscosity at 150 mg/mL and 20 °C (Table S2), as reported in Figure 1. One

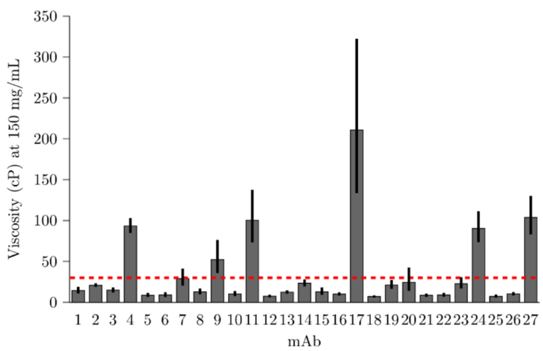


**Figure 1.** Fitted viscosity values of the 27 mAbs at 150 mg/mL. The red dashed line indicates the threshold value (30 cP) for high viscosity. There are 6 high-viscosity mAbs: mAb4, mAb9, mAb11, mAb17, mAb24, and mAb27. The error bars represent the 95% confidence interval.

can note that 21 mAbs have a low viscosity, defined as less than 30 cP.[14] Of these 21 mAbs, 8 have viscosities of 10 cP or less, with mAb18 having the lowest viscosity, 7.1 cP. Six mAbs (mAb4, mAb9, mAb11, mAb17, mAb24, and mAb27) exhibit high viscosities ranging from 52.3 to 210.9 cP. MAb9 is the only antibody with a $\lambda$ light chain. The other mAbs all have $\kappa$ light chains.

**3.2. Prediction of the Viscosity Behavior.** We began by attempting to predict antibody viscosity from the SCM score and the net antibody charge. The SCM score measures the spatial distribution of negative charges on the Fv region. It assumes that the negatively charged patches on the Fv region can induce strong attractive electrostatic interactions with positively charged residues, usually on the conserved regions of the mAb. The SCM scores above a cutoff value of 1000 are predicted to be high-viscosity mAbs. Figure 2A demonstrates the results predicted by the SCM score. All the high-viscosity mAbs are correctly classified. It is noted that there are eight false-positive cases that were predicted to be of high viscosity but were in fact of low viscosity. These mAbs contain 5 IgG1, 2 IgG2, and 1 IgG4 isotypes. The SCM score was developed and tested on mAbs of the IgG1 isotype, which have similar conserved regions. Overall charges on the conserved regions of the IgG1 isotype are positive at pH 6.0. Therefore, a negative charge patch on the Fv region is expected to be a good predictor to classify high-viscosity and low-viscosity mAbs.[16] Conversely, the charge distribution for IgG2 and IgG4 isotypes


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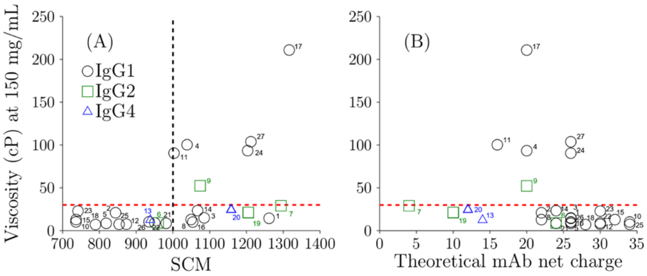


**Figure 2.** (A) Relationship of viscosity with SCM scores. The dashed line indicates the threshold SCM value (1000), above which high viscosity is expected. (B) Relationship of viscosity with theoretical mAb net charges based on the homology model. The predicted charges do not consider the effect of buffer ions present in solution and should be shifted to compare with an effective charge measured in the experiment. Labels are the mAb numbers corresponding to Figure 1. The red dashed line indicates the threshold value (30 cP).

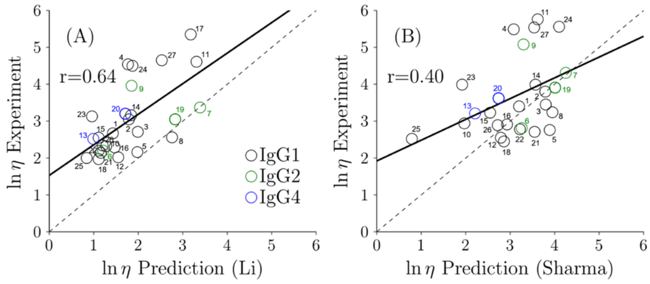


**Figure 3.** Comparison of the experimental viscosity ($\eta$) with two predictive models developed by Li et al.[17] and Sharma et al.,[10] respectively. The solid line is the least squares regression line. The dashed line is the identity line. Labels are the mAb numbers corresponding to Figure 1.

differs from those of IgG1s.[14,40] The conserved regions of IgG2s and IgG4s have fewer positively charged residues (mostly His, Arg, and Lys), which can lead to a different electrostatic behavior. However, even excluding all the studied IgG2 and IgG4 mAbs, there are still 5 out of 21 misclassified IgG1 mAbs. This suggests that the viscosity behavior is not determined solely by negatively charged patches on the Fv region.

To account for the charge difference between isotypes, we also plotted the viscosity with the net charges on the full-length mAbs, as shown in Figure 2B. The high-viscosity mAbs exhibit a wide distribution of net charges: the lowest net charge is +16 and the highest net charge is +26. All the mAbs with net charges below +14 and above +28 had low viscosity. It is likely that these mAbs exhibit different mechanistic behaviors that minimize solution viscosity.[14] For mAbs with high net charges, all domains, including Fv, CH1, CH2, and CH3, are positively charged inducing strong self-repulsions while mitigating reversible self-association. For mAbs with low net charges, the large number of negative charges in the Fv might lead to attraction that can result in tight packing, leading to large reversible cluster formation but with strong repulsion between reversible clusters. This will lead to high opalescent values as discussed elsewhere.[14] Finally, mAbs with net charge values in between 14 and 28 can exhibit high viscosity. These mAbs might exhibit attractive or weakly repulsive interactions that can result in extended network formation at high concentration (also called reversible self-association or extended cluster formation) that can lead to increased solution viscosity. However, only 6 of the 16 mAbs in this charge range exhibit high viscosity, indicating that the viscosity behavior is also governed by factors other than the net charge.

We have also compared the accuracy of the viscosity at high concentration predicted using other models. Figure 3 shows the results of viscosity models developed by Li et al.[17] and Sharma et al.,[10] respectively. The Li model was fitted by the viscosity at 150 mg/mL of 11 proprietary mAbs at pH 5.8 in 20 mM histidine-HCl buffer. This formulation condition is similar to that of the present study. Figure 3A reports that although there is a moderate correlation with the experimental data ($r = 0.64$), the Li model underestimates the experimental viscosity for most of the mAbs in this work. Most of the data are above the identity line, indicating that experimental viscosities are much higher, especially on a logarithm scale. The Sharma model predicts viscosity at 180 mg/mL, which is based on the Fv charge, the product of VH and VL charges, and the hydrophobicity index at a given pH. The viscosity at mAb concentrations of 180 mg/mL was determined using the exponential best fit line for the experimental data using eq 1. The Sharma model results in Figure 3B show a poor correlation with the experimental viscosity ($r = 0.40$). It is noted that the Sharma model was fitted with the viscosity data in a buffered solution at pH 5.5 and 200 mM arginine-HCl. The difference between the solution conditions may affect the performance of the model.

In addition, we applied Therapeutic Antibody Profiler (TAP) computational tool, which predicts antibody developability based on five metrics (Table S3).[38] Only one high-viscosity antibody (mAb17) had two amber flags. All other high-viscosity mAbs had only green flags. Additionally, five low-viscosity mAbs had one amber flag. Overall, the TAP analysis does not differentiate well between low- and high-viscosity mAb's in this dataset.

**3.3. Identification of the Key Determinants for the Viscosity Behavior from Machine Learning.** In the machine learning model, all 27 mAbs were partitioned into training and test sets for model evaluation and feature selection. Although as noted in a previous study, mAbs with high or low net charges can already be identified as having low viscosity due to charge effects,[14] it is unclear what the threshold values should be. Therefore, we attempted to keep all the data in the first round of model training and feature selection. The machine learning models include molecular descriptors such as charge and hydrophobic and hydrophilic properties in the variable regions and full-length antibodies (Table 1). In the case of highly correlated descriptors ($r \geq 0.95$), only one of the descriptors remained in the set. In total, 17 features remained after data preprocessing.

Table 2 lists the results for the area under the precision-recall curve (AUPRC) and accuracy of each feature averaged from 100 stratified threefold cross-validation runs using logistic regression. The accuracy is defined as the percentage of correctly predicted low- and high-viscosity data. Because there are many more low-viscosity antibodies than high-viscosity antibodies, the accuracy of the baseline model which predicts that all mAbs have low viscosity is 0.78. The baseline AUPRC is the fraction of mAbs with high-viscosity data, which is 0.22. The top three features are the number of hydrophobic or hydrophilic residues in the Fv regions and the CSP of the whole mAb. Figure 4 illustrates the relationship between viscosity and the top features. It is observed that the high-viscosity mAbs are generally rich in hydrophilic residues and

**Table 2. List of the Average Area under the Precision-Recall Curve (AUPRC) and Accuracy Scores with Their Standard Errors of the Mean for Each Feature by Threefold Cross-Validation**[a]

| feature | AUPRC | accuracy |
|---|---|---|
| *N*_phobic_Fv | 0.55 ± 0.03 | 0.86 ± 0.01 |
| *N*_philic_Fv | 0.51 ± 0.03 | 0.83 ± 0.01 |
| CSP_mAb | 0.41 ± 0.03 | 0.81 ± 0.01 |
| pI_Fv | 0.34 ± 0.02 | 0.75 ± 0.01 |
| SCM_Fv | 0.28 ± 0.02 | 0.74 ± 0.01 |
| SASA_philic_Fv | 0.25 ± 0.02 | 0.79 ± 0.01 |
| *N*_philic_mAb | 0.24 ± 0.01 | 0.75 ± 0.01 |
| SASA_phobic_mAb | 0.24 ± 0.01 | 0.77 ± 0.01 |
| CSP_Fv | 0.23 ± 0.01 | 0.76 ± 0.01 |
| SCM_mAb | 0.22 ± 0.01 | 0.74 ± 0.01 |
| SAP_Fv | 0.22 ± 0.01 | 0.77 ± 0.01 |
| HI_Fv | 0.22 ± 0.01 | 0.78 ± 0.01 |
| net charges_mAb | 0.22 ± 0.01 | 0.75 ± 0.01 |
| pI_mAb | 0.22 ± 0.01 | 0.74 ± 0.01 |
| SASA_philic_mAb | 0.22 ± 0.01 | 0.77 ± 0.01 |
| HI_mAb | 0.22 ± 0.01 | 0.78 ± 0.01 |
| SASA_phobic_Fv | 0.22 ± 0.01 | 0.78 ± 0.01 |

[a]The AUPRC is a useful measure when the datasets are very imbalanced. The baseline AUPRC and accuracy are 0.22 and 0.78, respectively.

poor in hydrophobic residues. This is in contrast to previous studies that hydrophobicity has a weak positive correlation with viscosity.[10,18] We will discuss this difference in more detail later. We also compared the results using the number of hydrophobic and hydrophilic residues in the CDR regions. The average AUPRC is 0.35 and 0.26, respectively. Using only the CDR regions as features to classify viscosities performed worse than that of the whole Fv regions. In the case of the mAbCSP feature, there were three high-viscosity mAbs with negative CSP values. This is due to the overall negative charges on the light chains. However, there were three other high-viscosity mAbs with positive CSP values, indicating that noncharge factors also govern the viscosity behavior.

Since the top one-feature combination was either the number of hydrophobic or hydrophilic residues in the Fv region, it is informative to analyze the amino acid composition in the Fv region for the low- and high-viscosity mAbs. Figure 5 reports the average number of each amino acid for the low- ($N$ = 21) and high ($N$ = 6)-viscosity mAbs in the Fv regions. For hydrophilic residues, the numbers of glutamic acid (E) and serine (S) are statistically significantly higher for the high-

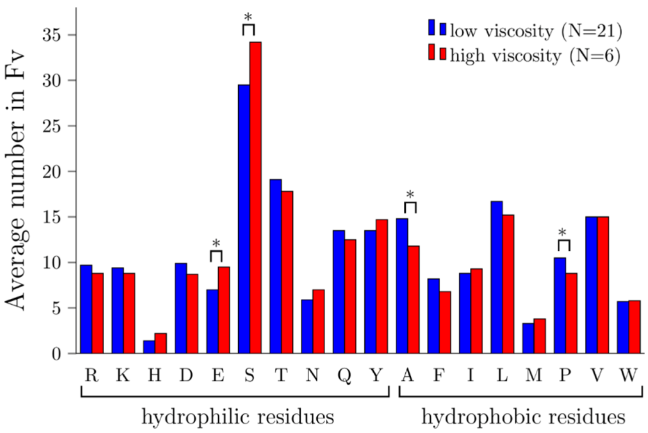


**Figure 5.** Average number of hydrophilic and hydrophobic residues in the Fv region for the low- and high-viscosity mAbs. The asterisk symbol indicates $p$-value < 0.05.

viscosity group. This indicates that the difference in the number of hydrophilic residues between low- and high-viscosity mAbs comes from both the charged residues and the polar uncharged residues. For hydrophobic residues, the numbers of alanine (A) and proline (P) are statistically significantly lower for the high-viscosity group. Figure S1 reports the average number of each amino acid in the CDR regions. Only the numbers of glutamic acid (E) and proline (P) are statistically significantly different. The $p$-value for serine is 0.06. Although the germline frameworks have high homology, they still have non-negligible effects on the amino acid compositions.

### 3.4. Feature Engineering and Model Development.

Feature engineering is a process to modify features or create features based on domain expertise for a particular application.[44] There are 21 low-viscosity mAbs in the dataset of 27 mAbs. Using only the single best feature leads to the correct classification of only 23 mAbs, which is only two more than the baseline model that predicts all mAbs have low viscosity. Based on the first-round selection, we aimed to improve our models by incorporating derived features and experimental observation. First, the high-viscosity mAbs are rich in the hydrophilic residues and poor in the hydrophobic residues in the Fv region; therefore, it is more convenient to define a new feature that combines both features. We propose an HVI for mAbs with the following formula

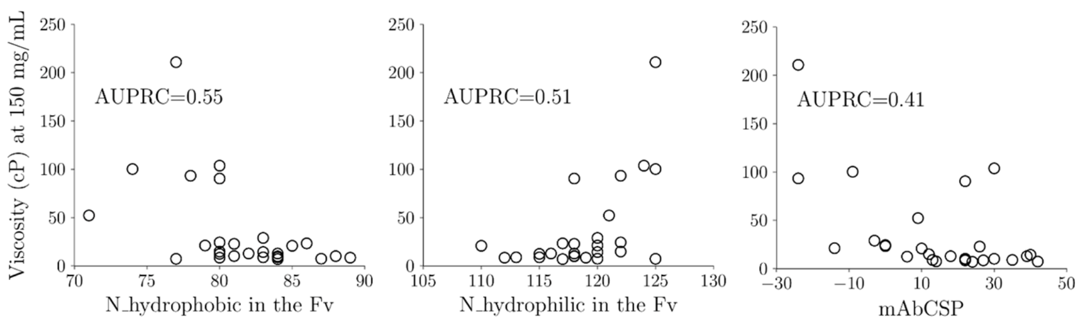


**Figure 4.** Relationship between viscosity and top three highest AUPRC features. $N$_hydrophobic and $N$_hydrophilic indicate the number of hydrophobic and the number of hydrophilic residues, respectively. CSP represents the charge symmetry parameter.


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$$\mathrm{HVI} = \frac{N\_\mathrm{hydrophilic\ Fv} - N\_\mathrm{hydrophobic\ Fv}}{\mathrm{Fv\ length}} \times 100\% \quad (2)$$

The difference in the residues is divided by the length of the Fv region and is represented as a percentage format. Figure 6A

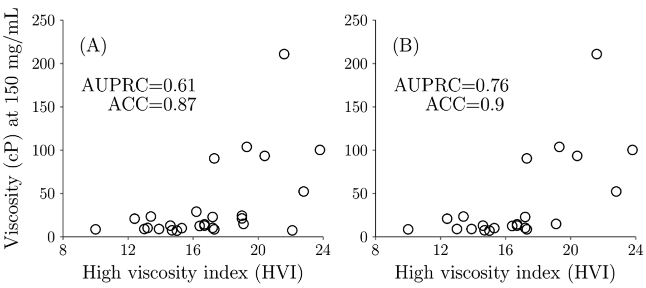


**Figure 6.** Relationship between viscosity and the HVI for (a) all mAbs and (b) mAbs in the middle charge group (between +12 and +34).

demonstrates the correlation between HVI and the viscosity behavior. The HVI feature further improves the AUPRC and accuracy compared to that using solely *N*_hydrophobic or *N*_hydrophilic in the Fv regions.

There are four mAbs (3, 19, 20, and 25) with HVI ≥19 that are false-positives (Figure 6A, Table S1). We found that three of them have either very high or very low mAb net charges. The charges for mAb19, mAb20, and mAb25 are +10, +12, and +34, respectively. From a recent study, it was found that mAbs with a high net charge tend to have low viscosity due to strong repulsion and mAbs with a low net charge tend to have high opalescence but low viscosity.[14] Our results confirmed that it is important to first filter out the mAbs with low viscosity due to charge effects. However, this nonlinear behavior was not captured by logistic regression algorithm. Therefore, we decided to apply a decision tree model that works better with nonlinear features. We aimed to find a lower bound and an upper bound for the mAb net charge for high-viscosity mAbs, with mAbs outside this range having low viscosity due to charge effects. From Figure 2B, the lower bound for the net charge can be anywhere from 12 to 16 and the upper bound can be anywhere from 28 to 34. We performed a grid search for the lower and upper bound within these ranges. For each pair, the remaining mAbs after the filtering process were trained using HVI as in Figure 6A, and the pair with the best AUPRC and accuracy was selected. We found that the lower bound is 12 and the upper bound is 34. Figure 6B demonstrates the results for using only the middle net charge group. The AUPRC was improved from 0.61 to 0.76 with the net charge filter. With these results, we proposed a decision tree model that combines net charge and HVI criteria (Figure 7) for classifying low- and high-viscosity mAbs. A concise mathematical form is also shown below.

$$\text{low viscosity} = \begin{cases} \text{net charges} \geq 34 \text{ or } \leq 12 \\ 12 < \text{net charges} < 34,\ \ \mathrm{HVI} < 17.3 \end{cases} \quad (3)$$

Table 3 shows the confusion matrix using the above-mentioned viscosity predictor. Twenty-six out of 27 mAbs are correctly classified. This expression facilitates fast viscosity screening for pharmaceutical development.

**Table 3. Confusion Matrix of the 27 mAbs in This Study Using the Decision Tree Model for Viscosity Prediction**

| | | predicted | |
|---|---|---|---|
| | viscosity | high | low |
| actual | high | 6 | 0 |
| | low | 1 | 20 |

**3.5. Applying the Predictive Models to New Antibodies.** The sequence information of high-viscosity mAbs is usually not available in the literature, preventing us from applying our decision tree model to the Tomar/Sharma mAbs. Instead, we applied a decision tree model using our approach to calculate the viscosity behavior of 38 antibody mutants recently reported by Apgar et al.[22] The authors performed two rounds of computational design to mitigate high concentration viscosity of a parental antibody. The viscosity was measured at pH 5.8 with histidine-sucrose as a diluent. This buffer condition was slightly different from that of our study. Figure 8A shows the net charge of all the mAbs in the Apgar study.

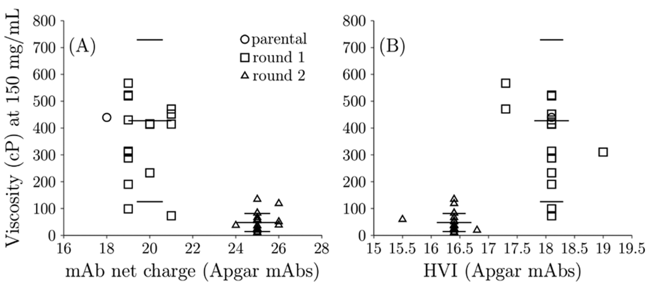


**Figure 8.** Relationship between viscosity for mAbs obtained from Apgar et al.[22] and (A) mAb net charge and (B) HVI. The cluster at 18.1 is majorly round 1 design (square), and the cluster at 16.4 is majorly round 2 design (triangle). The error bars indicate the average and standard deviation of the viscosity of round 1 and round 2 mAbs.

The parental antibody has the lowest net charge +18. The mutants have a net charge between +19 and +26. All these antibodies are within the middle charge group from our decision tree model; therefore, the viscosity prediction

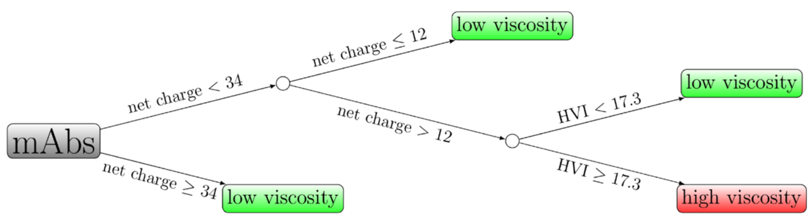


**Figure 7.** Decision tree model for classifying low- and high-viscosity mAbs based on mAbs net charge and the HVI.

depends on the HVI score. Notably, our decision tree model indicates that the net charges of the round 2 mutants are not high enough to prevent high viscosity. Figure 8B shows the prediction results of the HVI score (see the Supporting Information for details). There are two clusters corresponding to the round 1 and round 2 designs. The cluster with higher viscosity corresponds to the round 1 design, having a higher HVI score. The average viscosity was 422.7 (cP). The cluster with low viscosity corresponds to the round 2 design, having a lower HVI score. The average viscosity was 48.8 (cP). For the Apgar mAbs, the cutoff between the low-viscosity and high-viscosity clusters is between HVI scores of 17 and 18. This is consistent with our results. All the mAbs above the cutoff exhibited high viscosity; however, there were a couple of false-negative cases below the cutoff. Unlike our dataset which has a large gap between the low- and high-viscosity values, there are multiple Apgar mAb's with viscosity slightly above 30 cp. It is noted that some high concentration viscosities were obtained by extrapolation from low concentration viscosities for the Apgar mAbs, which might result in some uncertainties. To better evaluate the classification accuracy, we used different viscosity cutoffs. The classification accuracies using the decision tree model are 0.68, 0.79, and 0.87 using a cutoff value of 30, 40, and 60 cp, respectively. This shows that the experimental uncertainty should be considered when using the decision tree model.

The major difference between round 1 and round 2 antibodies includes four mutations on the light chain (E3V, S13A, Q16K, and S19R).[22] In addition to increased positive charges, there were two mutations from hydrophilic residues to hydrophobic residues. Interestingly, these two hydrophilic (E and S) residues were found to be rich in the high-viscosity mAbs in this study (Figure 5). The S19R mutation also replaced one more serine residue. Additionally, one of the low-viscosity mutants R2-006 (12.6 cP) contains another serine mutation (S51K). Whether the reduced viscosity comes from the removal of serine or addition of positive residues requires more study. In the round 1 design, all the mutants had increased positive charges or decreased negative charges; however, no serine mutations were included. Most of the round 1 mutants still exhibited high viscosity. This result supports that elevated viscosity could depend not only on the charged residues but also on other noncharged residues on the surface. Although our model was developed from 27 mAbs, it is able to predict the low/high viscosity of a dataset of 38 mutants with accuracies of 68−87%, depending on the cutoff value for the high-viscosity class. This demonstrates that the small dataset has a sufficient variability to allow the machine learning algorithm and predictive model to apply to new mAbs.

**3.6. Correlation between Viscosity and Hydrophobic/Hydrophilic Properties.** It was proposed that increased viscosity correlates with hydrophobicity, although to a lesser extent.[10,18] This is in contrast to our finding that high viscosity correlates with the number of hydrophilic residues. After comparing the antibodies used in previous studies with the current study, we found that the difference comes from limited high-viscosity data in previous studies. In Sharma's work, they measured viscosity of 14 mAbs at 180 mg/mL.[10] The highest viscosity was 80 cP. If we interpolate the viscosity using exponential fit assuming the viscosity is 1 cP at 0 mg/mL, the highest viscosity was only 38 cP at 150 mg/mL.[18] These 14 mAbs exhibited relatively stable viscosity behaviors. The weak correlation with hydrophobicity applies only to low-viscosity mAbs, and it may not be generalized to high-viscosity mAb's. From Figure 2B, the prediction from Sharma's model did not capture the high-viscosity behavior in this study. In Tomar's work, they measured viscosity of 16 mAbs and only 2 mAbs had high viscosity (>90 cP) at 150 mg/mL. They proposed that the hydrophobic surface area correlates weakly with the viscosity. Upon closer inspection, we observed that those high-viscosity mAbs could potentially be outliers (see the Supporting Information[18]). Taken together, the argument that hydrophobicity correlates with viscosity may be only applicable to low-viscosity mAbs. Our study nearly doubled the number of mAbs with more high-viscosity data available. We aimed to find a binary predictor to separate low- and high-viscosity mAbs. We found that hydrophilicity, rather than hydrophobicity, is a dominating factor for high-viscosity mAbs.

Weak interactions are crucial for protein−protein association and viscosity behaviors.[8] These interactions include hydrophobic and other hydrophilic charge−charge, charge−dipole, dipole−dipole, and hydrogen-bonding interactions.[45] In this work, we found that the high-viscosity mAb's have many serine residues in the Fv region (Figure 5). We hypothesize that the small size of serine could promote the formation of extensive hydrogen bonding between proteins mediated by solvents without much steric hindrance and loss of conformational entropy. Although the mechanistic study is beyond the scope of this work, the new insight from this study can guide future experimental design and model development to investigate antibody viscosity behavior.

## 4. CONCLUSIONS

In this work, we conducted concentration-dependent viscosity measurements of 27 FDA-approved antibody drugs. Six mAbs showed high viscosity in concentrated solutions. Based on the charge analysis, mAbs with high or low net charges generally have low viscosity due to strong repulsive interactions between proteins or strong attractive interactions that leads to larger cluster formation and opalescence. However, mAbs with net charges in the middle region, between +12 and +34, exhibit considerably different viscosity behaviors. For these mAbs, the effect of nonspecific short-range interactions becomes prominent. Machine learning reveals that the number of hydrophilic and hydrophobic residues in the Fv region highly correlates well with the viscosity behavior. This correlation can be further improved when the strong repulsive mAbs are excluded. Based on this correlation, we proposed a decision tree model based on the experiment and used a machine learning model to classify mAbs as having high or low viscosity. This model could serve as a tool to assess the risk of a high solution viscosity during the early stages of antibody screening. The design strategy is also summarized in Figure S2.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.molpharmaceut.0c01073.

Primary sequence for all mAbs in this work is in seq_vis_SI.fasta; feature values for machine learning are in features_values_SI.csv; and details for the case study using HVI are in mutants_SI.xlsx (ZIP)

Average number of hydrophilic and hydrophobic residues in the CDR region for the low- and high-viscosity mAbs; frame diagram of the experimental design and machine learning approach; 27 FDA-approved mAbs used in this study and their molecular properties and prediction scores; concentration-dependent viscosity measurement on the 27 mAbs used in this study; and five metrics of Therapeutic Antibody Profiler (TAP) and viscosity for the 27 mAbs in this study (PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

### Notes

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